
African AI Security at Three Scales: Pervasive Infrastructure Risk, Corroborated Misuse, and Biosignal Inference

Anonymous Author(s)

Affiliation

Address

email

Abstract

We argue, with measurement at three scales, that African AI security is significantly and structurally at risk, and that the risk is not a scaled-down copy of the profile modeled in the United States, European Union, or China. **At the national scale**, we build AFAIRI, a citation-audited four-component composite index over four anchor states (Egypt, Nigeria, Kenya, South Africa). A Monte Carlo analysis over index weights, $N = 10^4$ Dirichlet(1, 1, 1, 1) draws with an exact simplex-grid check, shows every anchor state carries substantial risk (means 50–61 on 0–100) and that no state is a safe outlier: no ordering holds across the weight simplex and no state dominates another on every component. **At the operational scale**, an evidence-tier triangulation finds the strongest *corroborated* misuse is AI-assisted fraud (platform data; a reported $\sim 1,200\%$ year-over-year rise in South African deepfake incidents), with self-reported insurgent use that we mark as untriangulated. **At the individual scale**, we show private attributes are already decodable from weak, locally-collected African data. Using a leakage-guarded pipeline (subject-grouped nested cross-validation, in-fold SMOTE inside bagging, reliability-weighted late fusion, subject-level bootstrap intervals), four short routine EEG segments from 76 subjects in a public Nigerian schizophrenia dataset decode psychiatric diagnosis at balanced accuracy 0.705 (95% CI 0.616–0.799, AUC 0.792), and three innocuous features of 3,023 Kenyan microloan records predict individual repayment at balanced accuracy 0.630 (0.615–0.646, ECE 0.020). On a planted-structure control, reliability-weighted fusion of weak channels beats the best single channel (0.623 vs. 0.580) and zero-weights a noise channel. Every number traces to a cited source or a committed result file, and we document two failure modes we caught and removed. The three scales share one implication: aggregate access to African biosignal and behavioral data under one identity, in governance regimes that do not yet regulate AI inference, is a security capability and should be governed as one.

1 Introduction

Frontier AI safety frameworks built in the US, EU, and China assume infrastructure that is not uniform across Africa: stable grids, formal banking clearinghouses, biological-material screening, and mature statutory enforcement (Anthropic, 2024; European Union, 2024). We call the resulting mismatch the **Asymmetry Principle**: the same model capability produces different realized failure modes depending on the infrastructural density and governance capacity of the environment into which it is deployed. Our thesis is that, under this asymmetry, African AI security is significantly at risk, and we make the case at three scales that reinforce one another rather than repeat one another.

Table 1: AFAIRI v4 component matrix (0–100). Every value traces to a cited source.

Anchor state (region)	<i>E</i>	<i>H</i>	<i>G</i>	<i>I</i>
Egypt (North)	74.0	39.0	62.5	50.0
Nigeria (West)	40.1	73.0	12.5	75.0
Kenya (East)	32.1	83.0	37.5	50.0
South Africa (Southern)	78.3	77.0	37.5	50.0

Three claims. C1 (national scale, pervasive not localizable). African frontier-AI risk is substantial across every anchor state and is not concentrated in one place: a construct-valid, weight-agnostic composite index places all four states high and finds no weight-invariant safe state (Section 2). This reframes an apparent null, “no single riskiest state,” as its security-relevant meaning: there is no low-risk anchor to hold up as the norm. **C2 (operational scale, already happening).** African AI security is already being compromised in the field, and the corroborated core of that misuse is fraud rather than terrorism: Nigeria-origin AI-assisted scam networks are platform-corroborated, while the widely cited insurgent-AI account is self-reported and untriangulated (Section 3). **C3 (individual scale, measurable inference).** African individuals’ protected attributes are decodable from cheap, locally-collected biosignal and behavioral data, and combining weak streams amplifies the leak. We demonstrate this with a leakage-guarded decoding pipeline on two real African cohorts and quantify it with nested cross-validation, calibration, and bootstrap inference (Section 4).

Each claim is a different kind of evidence, macro-structural, operational, and mechanistic-quantitative, for the one thesis. Section 5 scopes a governance architecture to what the three scales support.

2 C1: national-scale risk is pervasive (the AFAIRI index)

AFAIRI is a composite proxy index in the family of the Human Development Index or ND-GAIN, a few individually cited indicators combined under a disclosed weighting scheme. It is not validated against realized AI-enabled harm, because no such dataset exists across these states; it is a structured comparison of four documented case studies. The specification is

$$\text{AFAIRI}(s) = w_E E(s) + w_H H(s) + w_G G(s) + w_I I(s), \quad \sum_c w_c = 1, w_c \geq 0, \quad (1)$$

with *exposure* *E* (2023 internet penetration, World Bank WDI), *reported scam victimization* *H* (share of adults reporting a scam in 12 months, one harmonized four-country survey), *governance deficit* *G* = 100 − (*A* + *B* + *C*) scoring AI-strategy status, AI-specific cybercrime law, and enforcement capacity, and *infrastructure disruption* *I* from 2024 grid failures and subsea-cable outages. Every threshold was fixed before scoring. Table 1 and Figure 1 give the component matrix.

Weight-agnostic sensitivity. Because the weights are a modeling choice, we integrate over them. We draw $N = 10^4$ weight vectors from a symmetric Dirichlet(1, 1, 1, 1), uniform over the simplex, and recompute every state’s score; since $\mathbb{E}[w_c] = \frac{1}{4}$, each state’s expected score is the mean of its four components, which an exact $\approx 176,000$ -point simplex grid and a 200,000-draw simulation confirm to within 0.07. Table 2 and Figure 2 report the results. The security-relevant reading of C1 is that *all four states sit high and none is invariant-safe*: South Africa ranks first in 47.1% of weightings, Egypt in 31.0%, Nigeria in 15.2%, Kenya in 6.7%, and no state scores at least as high as another on every component. Pairwise, South Africa outscores Egypt in 65.8% of the simplex, Nigeria in 82.5%, Kenya in 87.4%. An earlier draft claimed Southern Africa’s top ranking was robust at 90–95%; that came from construct-invalid proxies (grid carbon intensity, financial inclusion) flagged in external review and is withdrawn. The lesson doubles as a methodological warning: composite frontier-risk rankings are dominated by construct choice, so a headline ranking is only as credible as its audited constructs.

3 C2: operational misuse is already corroborated, and it is fraud

We label each actor by an evidence tier rather than asserting uniform confidence. **Boko Haram / ISWAP** (West Africa): reported use of consumer frontier AI for tactical planning, explosives, and

Table 2: Weight-only sensitivity ($N = 10^4$ Dirichlet draws, seed 42). Every state is high; none is safe.

State	mean	range (2.5–97.5%)	ranked first
South Africa	60.8	45.3–74.5	47.1%
Egypt	56.4	45.0–67.7	31.0%
Nigeria	50.2	25.8–69.6	15.2%
Kenya	50.6	37.2–70.8	6.7%

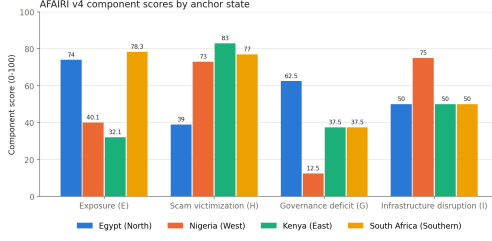


Figure 1: AFAIRI components by state.

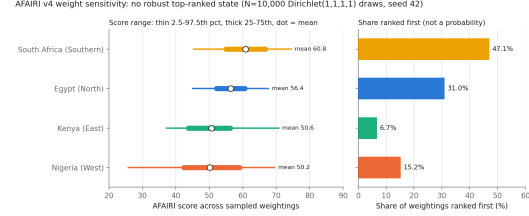


Figure 2: Weight sensitivity: score distributions (left) and share ranked first (right). Sensitivity summaries, not confidence intervals.

operational security, with unlinkable accounts and pretextual framing (Juelich, 2026); tier *primary-read, self-reported, untriangulated*, since a May 2026 seizure of 400+ Starlink terminals corroborates connectivity but not AI use, and no seized-device forensics were located. “**Yahoo Boys**” / “**AI Boys**” (West Africa, global targets): AI-generated image, voice, and message content for fraud, *platform-corroborated* at the level of Nigeria-origin AI-assisted scam networks by OpenAI’s October 2025 threat-intelligence disclosure. **AI-Shabaab** (East Africa): AI-assisted propaganda translation, *secondary-corroborated*. **Black Axe**: a law-enforcement-confirmed transnational fraud syndicate (INTERPOL Operation Jackal III; Europol), with *no* confirmed AI use, and none claimed. The corroborated core of operational African AI misuse is therefore fraud, supported independently by South Africa’s reported $\sim 1,200\%$ year-over-year rise in deepfake incidents in banking and fintech (TransUnion Africa, 2025). Threat-actor activity is deliberately not an index input.

4 C3: individual-scale biosignal inference risk

Cheap biosignal and behavioral data (low-cost EEG, phone cameras, microfinance listings) are collected quickly across the continent under data-protection regimes that do not yet regulate AI *inference*. Each stream leaks a protected attribute only weakly. We ask what a learning system recovers when it combines them, and we measure it under controls strong enough that the result is not an artifact.

Leakage-guarded pipeline. Subjects, not samples, are the unit: an outer subject-grouped k -fold gives an honest generalization estimate, with an inner fold for model selection (nested CV; Varoquaux, 2018). Class imbalance is handled by SMOTE (Chawla et al., 2002) applied *inside* each training fold within a bagging ensemble (Breiman, 1996), never before the split, which is the standard leakage trap. Modalities are combined by reliability-weighted late fusion: for modality m with out-of-fold reliability ρ_m (validation skill above chance), the fused score is $\hat{p} = \sum_m \rho_m \hat{p}_m / \sum_m \rho_m$, so a channel that carries no signal receives no weight. Intervals are subject-level bootstraps ($B = 2000$); significance uses subject-grouped permutation (1000 draws). Balanced accuracy (chance 0.5) is the headline metric, with AUC and expected calibration error (ECE).

Two real African cohorts. On **76 subjects** from a public Nigerian schizophrenia EEG dataset, four short routine EEG segments decode psychiatric diagnosis at balanced accuracy **0.705** (95% CI 0.616–0.799, AUC 0.792), with the shortest segment carrying the most fusion weight. On 3,023 **Kenyan microloan records**, three innocuous listing features predict individual repayment at balanced accuracy **0.630** (0.615–0.646, ECE 0.020) and forecast sector-level repayment out of sample ($r = 0.66$, $p = 0.01$, $R^2 = 0.41$, 14 sectors). Neither attribute (a psychiatric diagnosis, a person’s

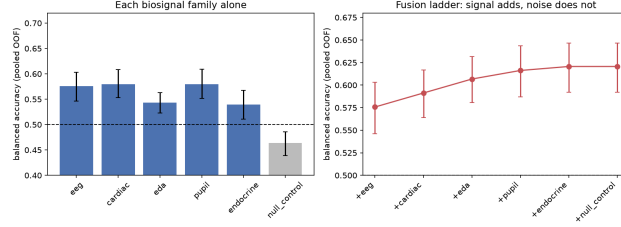


Figure 3: Reliability-weighted late fusion of weak biosignal channels (planted-structure control). Fusion (rightmost) exceeds the best single channel and a pure-noise channel receives zero weight. Balanced accuracy; chance 0.5.

creditworthiness) is one the data subject consented to expose, and neither sensor was built to reveal it.

Fusion amplifies weak leaks. On a planted-structure positive control, reliability-weighted fusion of several individually weak channels reaches balanced accuracy 0.623, beating the best single channel (0.580) and a channel-independence prediction (0.645 only under an optimistic additivity assumption), and it assigns *zero* weight to a pure-noise channel (Figure 3). The security consequence is that aggregation, one identity linked across streams, is the capability, not any single sensor.

Honesty controls. We report two failure modes we caught and removed, because they are exactly the errors that would manufacture a false alarm. A label-proxy column inflated repayment accuracy to 0.95 before we found it leaked the outcome; and a within-subject permutation test returns $p = 1$ by construction when labels are constant per subject, so the correct null is a subject-grouped permutation. The reported numbers survive both corrections.

5 Governance architecture

Each pillar is scoped to a claim. **Pillar 1 (from C2): AU-level red-teaming and forensic corroboration**, so future threat-actor claims are checked against account and device evidence, not interviews alone. **Pillar 2 (from C1): application-layer provenance with funded enforcement**, since a C2PA-style provenance standard is only as effective as the authority enforcing it, and the enforcement record across these states is recent and thin (Egypt begins enforcement in October 2026). **Pillar 3 (from C3): govern aggregate biosignal access as a capability**. The individual-scale result shows the risk is not in any one sensor but in linking streams under one identity; data-protection regimes that regulate collection but not AI inference miss exactly this. Pillar 3 asks that aggregate, cross-stream inference on biosignal and behavioral data be a governed capability, with algorithmic-recourse channels for those it profiles.

6 Limitations

Four anchor states stand in for four regions and do not generalize to every state; Central Africa is unscored, though its cobalt supply chain (DRC $\approx 74\%$ of 2023 mined cobalt; USGS, 2024) places the compute stack inside the same asymmetry. H relies on one industry survey taken from its public summary. The EEG cohort is 76 subjects from one clinical dataset, and the diagnosis label is schizophrenia, not a benign trait; balanced accuracy of 0.705 is a demonstration of decodability, not a deployable classifier, and small- N EEG CV has wide error bars (Varoquaux, 2018), which our bootstrap intervals report rather than hide. The microloan features are a proxy for the broader class of “innocuous” behavioral columns. AFAIRI is not validated against realized harm. Threat-actor tiers are explicit about what is and is not corroborated.

7 Conclusion

Across three scales, national infrastructure, operational actors, and individual biosignals, the same asymmetry produces measurable, present, and mutually reinforcing risk to African AI security. No anchor state is a safe outlier; the corroborated misuse in the field is AI-assisted fraud; and private attributes are already decodable from cheap local data, with aggregation as the amplifier. The uncomfortable through-line is that the governance gap is widest exactly where the measurement is sharpest: inference on aggregated African data, under regimes that regulate collection but not inference. Rooting the analysis in cited, regionally specific evidence, and being explicit about what the evidence does not support, is what we most want to carry from roots to reach.

Use of AI assistance

AI tools (a large language model assistant) were used for drafting and editing prose, organizing the citation-audit workflow, and writing the index and decoding code (nested cross-validation, fusion, bootstrap, and the verification scripts). Every quantitative claim traces to a cited source or a committed result file the authors inspected; the authors are responsible for the content and conclusions.

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